A Multi-Modular Fusion Framework for Occlusion-Resistant Pediatric Patient Counting:

Synthesizing Part Affinity Fields, Projective Geometry, Allometric Canons, and Anchor-Free Tracking

Research Framework Synthesis

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Abstract

Standard computer vision algorithms consistently fail in healthcare environments where cultural practices dictate severe occlusion (e.g., infants carried on mothers' backs). This document presents a comprehensive fusion of four distinct computational methodologies to solve this ”Invisible Child” anomaly. By synthesizing Part Affinity Fields (PAFs) for detection, Inverse Perspective Mapping for spatial correction, Anthropometric Canons for biological classification, and Anchor-Free Kinematics for trajectory tracking, we establish a mathematically rigorous pipeline. This paper deconstructs the underlying mathematics of each module term-by-term, proving how the sequential fusion of these algorithms yields an occlusion-resistant, privacy-preserving pediatric counting system.

1 Introduction: The Sequential Necessity

The architectural failure of ”off-the-shelf” object detection (e.g., standard Haar Cascades or naive YOLO bounding boxes) stems from their reliance on 2D pixel textures and Non-Maximum Suppression (NMS). When a child is bound to an adult's back, the semantic features merge, forcing the algorithm to delete the child's bounding box.

To resolve this, we propose a pipeline that physically separates the act of *detecting humans* from *measuring them*. The pipeline operates in four deterministic modules:

1. **Module 1 (Topological Detection):** Isolating overlapping skeletons using Vector Fields (Cao et al., 2017).
2. **Module 2 (Spatial Correction):** Eliminating CCTV perspective distortion using Projective Geometry (Tosti et al., 2021).
3. **Module 3 (Biological Classification):** Distinguishing children from adults using the Allometric Canon (Ince et al., 2014; Domljan et al., 2024).
4. **Module 4 (Identity Tracking):** Preventing double-counting via Anchor-Free Kinematics (Zhang et al., 2021).

2 Module 1: Biomechanical Detection via Part Affinity Fields (PAFs)

2.1 The Premise

Standard bounding boxes fail because the intersection over union (IoU) of a carried baby and a mother approaches 1.0. Cao et al. (2017) proved that we must bypass bounding boxes entirely and track topological joints using a ”bottom-up” approach called Part Affinity Fields (PAFs).

2.2 The Mathematics of PAFs

A PAF is a 2D vector field that encodes the position and orientation of human limbs over the image domain. Instead of asking "Is there a human here?", PAFs ask "Does this elbow belong to this shoulder?"

If a person k is in the image, and we are evaluating a limb c that connects joint j_1 and joint j_2, the algorithm defines an ideal vector $\mathbf{v} = (j_2 - j_1)/\|j_2 - j_1\|$. The PAF at any pixel $\mathbf{p}$ is defined as:

$$\mathbf{L}_{c,k}(\mathbf{p}) = \begin{cases} \mathbf{v} & \text{if } \mathbf{p} \text{ lies on limb } c \\ 0 & \text{otherwise} \end{cases} \tag{1}$$

To determine if two detected joints (e.g., a mother's torso and a smaller head) actually belong to the same person, the algorithm calculates the line integral over the line segment connecting them:

$$E = \int_0^1 \mathbf{L}_c(\mathbf{p}(t)) \cdot \mathbf{v}\, dt \tag{2}$$

Term-by-Term Breakdown:

- $\mathbf{L}_c(\mathbf{p}(t))$: The predicted Part Affinity Field evaluated at a specific pixel **p** along the line t.

- **v**: The normalized directional vector between the two detected joints.

- E: The confidence score (Affinity).

Synthesis Impact: If a baby is on a mother's back, calculating E between the mother's shoulder and the baby's neck will yield a near-zero or negative score because the vector flow violates human biomechanics. The algorithm mathematically rejects the connection, successfully splitting the single pixel mass into **two distinct skeletons** (Mother and Child).

3 Module 2: Spatial Correction via Projective Geometry

3.1 The Premise

Once Module 1 provides the raw 2D pixel coordinates of the baby's head and body, we encounter the perspective distortion of ceiling-mounted CCTV cameras. As Tosti et al. (2021) demonstrated in forensic metrology, raw pixels (u, v) cannot be used to calculate biological ratios because objects further from the lens appear disproportionately smaller.

3.2 The Mathematics of the Homography Matrix

To un-tilt the camera, we assume the hospital floor is a flat ground plane ($Z = 0$). We construct a Homography Matrix (H) that maps a true 3D coordinate in the room (X, Y) to the distorted 2D pixel on the screen (u, v).

To fix the distortion, we multiply the joint coordinates provided by Module 1 by the **Inverse Homography Matrix** (H^{-1}):

$$s \begin{bmatrix} X_{true} \\ Y_{true} \\ 1 \end{bmatrix} = H^{-1} \begin{bmatrix} u \\ v \\ 1 \end{bmatrix} \tag{3}$$

Alternatively, to bypass full 3D reconstruction, we utilize the **Cross-Ratio (CR)**, which is strictly invariant under projective transformations. Given four collinear points (Top of Head T, Bottom of Neck N, Feet F, and the vertical Vanishing Point V):

$$CR(T, N, F, V) = \frac{d(T, F) \cdot d(N, V)}{d(N, F) \cdot d(T, V)} \tag{4}$$

Term-by-Term Breakdown:

- $d(T, F)$: The distorted pixel distance from the top of the head to the feet.
- V: The vanishing point (calculated once during camera calibration).
- CR: A constant mathematical value that remains identical regardless of camera pitch, yaw, or patient distance.

Synthesis Impact: The spatial coordinates are now perfectly flattened. Module 2 hands Module 3 true orthogonal measurements, totally immune to camera angle.

4 Module 3: Biological Classification via the Allometric Canon

4.1 The Premise

With isolated (Module 1) and distortion-free (Module 2) measurements, we must classify the age of the patient. Ince et al. (2014) proposed relative measurement ($r = l_H/l_B$) to differentiate children from adults. Domljan et al. (2024) provided the rigorous, contemporary statistical ground truth for this metric, defining the exact biological canons (C) of modern children.

4.2 The Mathematics of Allometric Scaling

Rather than utilizing heavy facial-recognition convolutions, we execute a simple, low-latency algebraic calculation.

$$C = \frac{BH}{HL} \tag{5}$$

Term-by-Term Breakdown:

- BH: Total Body Height (or Torso Height in severe occlusion), provided by the corrected output of Module 2.
- HL: Head Length, the vertical Euclidean distance between the vertex (V) and gnathion (gn).
- C: The Allometric Canon constant.

Based on Domljan et al.'s statistical distributions (where $p < 0.05$ confirmed significant canonical shifts across ages), we establish rigid thresholds:

- If $C \in [5.3, 6.2]$, the subject is classified as a **Toddler/Child** (Ages 2–6).
- If $C \geq 7.5$, the subject is classified as an **Adult**.

Synthesis Impact: The system successfully overrides the assumption that a smaller secondary box on an adult's back is a backpack. The geometric ratio mathematically proves the object possesses human pediatric proportions, registering a successful "count."

5 Module 4: Identity Tracking via Anchor-Free Kinematics (FairMOT)

5.1 The Premise

If the mother turns around, the baby's skeleton is completely obscured. If the camera loses the ID, it will recount the baby when the mother turns back around. Zhang et al. (2021) solved this via FairMOT, shifting away from bounding-box tracking to tracking the precise center-point of the person, coupled with a Kalman Filter.

5.2 The Mathematics of Spatio-Temporal Tracking

FairMOT estimates the exact center of the child using a Gaussian heatmap. The ground-truth center (cx, cy) is mapped to the heatmap Y using:

$$\hat{Y}_{x,y} = \exp\left(-\frac{(x-cx)^2 + (y-cy)^2}{2\sigma_p^2}\right) \tag{6}$$

To remember the child when they disappear, we pass their coordinates into a **Kalman Filter**, which uses a state vector $\mathbf{x} = [u, v, \gamma, h, \dot{u}, \dot{v}, \dot{\gamma}, \dot{h}]^T$. The filter updates its memory using the state transition equation:

$$\mathbf{x}_k = A\mathbf{x}_{k-1} + B\mathbf{u}_k + \mathbf{w}_k \tag{7}$$

Term-by-Term Breakdown:

- $\hat{Y}_{x,y}$: The probability that a pixel at (x, y) is the absolute center of the child.
- σ_p^2: The spatial variance (size) of the child.
- $\mathbf{x}_k$: The current predicted state of the hidden child at time k.
- $A\mathbf{x}_{k-1}$: The transition matrix predicting where the child is moving based on their velocity before they were hidden.

Synthesis Impact: The child's verified identity (from Module 3) is locked to a center-point (from Module 4). Even if the child is visually occluded for several seconds, the Kalman filter mathematically "ghost tracks" their trajectory along with the mother's movement, preventing duplicate counting upon re-emergence.

6 Conclusion: The Complete Fusion

By mathematically fusing these four distinct algorithms, we engineer a robust framework immune to the classical pitfalls of object detection.

The process is strictly linear and deterministic: **PAFs** break apart the merged pixels, **Projective Geometry** normalizes the space, **Allometry** provides the biological verification of age, and **Kalman filtering** guarantees the temporal integrity of the data. This synthesis provides a computationally lightweight, highly accurate solution capable of uncovering the "Invisible Child" in complex healthcare environments.